BURHAN KARAHASAN

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EDUCATION

Koç University, Istanbul. CGPA : 4.00 2024 – Present
Master of Science in Mechanical Engineering
Koç University, Istanbul. CGPA : 3.95, Summa Cum Laude 2018 – 2024
Bachelor of Science in Mechanical Engineering *(Third Rank)*
Bachelor of Science in Electrical and Electronics Engineering *(Double Major)*

RESEARCH EXPERIENCE

Koç University Medical Robotics Laboratory (Prof. Metin Sitti) 2024 – Present
Graduate Researcher *Istanbul, Türkiye*

- **X-ray image-guided control of a soft magnetic millirobot.** Built the platform end to end inside a cabinet X-ray system: motorized stages and an actuated end-effector, live image capture from the X-ray unit, calibration, and real-time control software. Now moving the platform to a professional robotic manipulator.
- Designed a finite-state-machine controller that commands the stepper-driven stages in small position steps, chosen over PID because USB latency and stepper actuation make continuous feedback unreliable; it gives stable, precise closed-loop navigation.
- Adapted an existing neural-network detector from prior group work (C. Wang et al., *Nature Machine Intelligence*, 2026) and integrated it into the real-time loop to locate the robot in low-contrast X-ray frames.
- Video: youtu.be/_P9mQ2cJjDY
- **Ongoing: physics-based modeling and design of soft magnetic millirobots.** Developing models of untethered soft magnetic millirobots to design and control the robots I build; manuscript in preparation, details withheld for now.

Koç University Manufacturing and Automation Research Center (Prof. Ismail Lazoglu) 2021 – 2024
Undergraduate Research Assistant *Istanbul, Türkiye*

- Robot-guided surgery: surgical path optimization avoiding critical zones and brain vessels.
- Mechanical ventilation device: worked on the serial communication of several sensors & devices, optimized the system and improved the sampling rate of sensors 20 times.

PUBLICATIONS & PRESENTATIONS

Peer-reviewed journal article
Munam Arshad, Eda Guven, **Burhan Karahasan**, Ismail Lazoglu. "A novel real-time wireless sensor integration for enhancing positive pressure system operation in single limb passive vented circuit." *Biomedical Signal Processing and Control* 85 (2023).

Manuscript under review
Burhan Karahasan, Ismail Lazoglu, Ihsan Solaroglu. "Path optimization for robotic surgery avoiding functional zones" (full title withheld pending review). Submitted to *Computer Methods and Programs in Biomedicine*; under review, not yet accepted.

Conference presentation (poster)
Burhan Karahasan, Ömer Yiğit, Chunxiang Wang, Metin Sitti. "Image-Guided Small-Scale Magnetic Robot Navigation Under X-Ray." Poster, 7th Turkish Robotics Conference (national), Koç University, İstanbul, July 2026.

Manuscripts in preparation (not yet submitted)
Burhan Karahasan et al. Physics-based modeling of soft magnetic millirobots (title withheld).
Burhan Karahasan et al. Journal manuscript extending the X-ray navigation work above (title withheld).

TECHNICAL SKILLS

Programming	MATLAB, Python, Java/JavaFX, C/C++, VHDL, Visual Basic.
Platforms	Siemens NX, SolidWorks, Fusion 360, Ansys Mechanical, Ansys Fluent.
Miscellaneous	Robotics, Machine Learning/Deep Learning, Computer Vision, Finite Element Analysis, Computer Aided Design/Manufacturing, Numerical Methods, Generative Design, LaTeX.
Languages	English (full professional; all degrees taught in English), Chinese (fundamental).

ACHIEVEMENTS

Elected Presidential Fellow of Koç University (2024).
Completed the track program in artificial intelligence.
Received Vehbi Koç Honor awards for 6 semesters.
DEVAK Scholarship recipient. Full merit scholarship recipient for the undergraduate education in Koç University.

INDUSTRY EXPERIENCE

Ubicro Aug. 2022 – Dec. 2022
Prototyping and Testing Engineer *Istanbul, Türkiye*

· Implemented electronic controllers on aeroponic farming machines and assembled prototypes.

DeltaV Space Technologies Aug. 2022 – Sep. 2022
Guidance, Navigation and Control Intern *Istanbul, Türkiye*

· Worked on flight data storage using STM32 microcontroller devices.
· Reported several research about deep learning applications on rocket systems, hybrid altimeter systems in supersonic aircrafts and calibration & characterization of IMU sensors & noise effects.

Turkish Aerospace Industries June 2021 – July 2021
Integrated Logistics Support Intern *Ankara, Türkiye*

· Worked with the equip that conducts tests on unmanned aerial vehicle engines and prepares the repair and maintenance manuals for post-sales in integrated logistics support department.
· Coded a form application that helps to filter the maintenance manuals.

TEAM PROJECTS

Koç University Autonomous Drone Team (KUADRONE) 2020 – 2021
Team Leader *Istanbul, Türkiye*

· Competed in TÜBİTAK's 5th International Unmanned Aerial Vehicle Competition.
· Organized the team hierarchy, and the documentation of the project.
· Devised the mission planning, thrust calculations and electronic circuit design.

RELEVANT COURSES

Robotics Introduction to machine learning Computer vision with deep learning Linear systems theory

TEACHING EXPERIENCE

Teaching assistant and peer academic support for students in MATH106: Calculus I, MECH201: Statics and Mechanics, MECH206: Dynamics, MECH304: Dynamical Modeling and Control, MECH491: Senior Design Project.

OBJECTIVE

Robotics researcher with end-to-end experience across robot hardware, imaging, real-time control, and physics-based modeling of soft robots. I aim to pursue a PhD in the control and autonomy of small-scale and medical robots.

REFERENCES

Prof. Metin Sitti — M.Sc. advisor; President, Koç University — msitti@ku.edu.tr
Prof. Ismail Lazoglu — Undergraduate research advisor, Koç University — ilazoglu@ku.edu.tr